

DHRUV MANDAL

Artificial Intelligence & Data Science Engineer

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PROFESSIONAL SUMMARY

B.Tech student in Artificial Intelligence & Data Science with hands-on experience building NLP classification pipelines, predictive machine learning models, and LLM-driven automation workflows in Python. Skilled in deploying containerized ML microservices with FastAPI and Docker, and in constructing end-to-end ML lifecycle pipelines with MLflow, Airflow, and SHAP explainability. Comfortable across the stack, from model fine-tuning with PyTorch and Hugging Face Transformers to front-end delivery with React.js and Next.js. Seeking internship or entry-level opportunities in AI, Machine Learning, or Data Science.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript/TypeScript (ES6+), HTML5, CSS3

AI & Machine Learning: Machine Learning, Deep Learning, NLP Pipelines, Predictive Modeling, LLM Fine-Tuning, RAG Systems, Explainable AI (SHAP)

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, XGBoost

Backend & APIs: FastAPI, REST APIs

Databases: SQL, MySQL

MLOps & Tools: Docker, MLflow, Apache Airflow, Git, GitHub, Linux, Jupyter, VS Code

Web Development: React.js, Next.js, Vite, Responsive Design, Front-End Architecture

Core Concepts: Data Structures, Object-Oriented Programming

EXPERIENCE

Machine Learning Developer & Automation Engineer

Jan 2026 – Jun 2026

O3GenAI IT Solutions

- Architected and deployed an end-to-end NLP classification pipeline in Python that automated triage and tagging for 2,000+ monthly customer tickets.
- Developed high-throughput lead intelligence and automation workflows leveraging the GPT-4 API and Playwright.
- Optimized prompt engineering and inference latency, cutting token consumption by 35%.

PROJECTS

Lead Scoring Engine

Python | XGBoost | FastAPI | Docker

- Trained a gradient-boosted (XGBoost) classification model on 18 months of CRM interaction data to rank leads by conversion probability.
- Deployed the model as a containerized REST microservice on FastAPI, reducing sales team outreach overhead by ~30%.

Brand Sentiment Monitor

Hugging Face | DistilBERT | Next.js | WebSockets

- Built a real-time streaming analytics dashboard classifying live social media streams at 500+ items/min.
- Fine-tuned a lightweight DistilBERT model, achieving sub-50ms inference time with 92% sentiment classification accuracy.

Clinical Medical Document Classifier

BioBERT | PyTorch | Scikit-learn

- Fine-tuned BioBERT on 12,000 clinical medical records to categorize documents into 14 ICD-10 diagnostic categories.
- Achieved a 91% macro F1 score on the validation set, outperforming a traditional TF-IDF baseline by 18 points.

End-to-End Churn Prediction Pipeline

MLflow | Airflow | SHAP | Python

- Constructed a complete ML lifecycle pipeline, engineering 40+ customer behavioral features with scheduled Airflow DAGs.
- Integrated SHAP explainability values into executive dashboards, providing actionable retention triggers.

EDUCATION

B.Tech in Artificial Intelligence & Data Science

2024 – Present

Ajeenkya D. Y. Patil School of Engineering, Pune | GPA: 8.7/10

Relevant Coursework: Deep Learning, Statistical Machine Learning, Natural Language Processing, Data Engineering

Higher Secondary Certificate (HSC – Science, PCM)

Jul 2022 – Mar 2024

Mother Mary's English High School and Junior College, Maharashtra State Board | Percentage: 82%

CERTIFICATIONS

- Microsoft Azure Fundamentals (AZ-900) — Microsoft
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — Oracle
- Oracle Data Platform 2025 Certified Foundations Associate — Oracle